

CompTIA

A+ Core 1

Exam 220-1201

A+ 220 1201

Module 7



Supporting Network Services

Module Outcomes



Summarizing services
provided by networked
hosts



Comparing Internet and
embedded appliances



Troubleshooting networks

Lesson 7.1

Networked Host Services

A+ 220-1201 Certification Exam Objectives:

- 2.1 Compare and contrast Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) ports, protocols, and their purpose.
- 2.3 Summarize services provided by networked hosts.



What services are commonly found on modern networks?



What port numbers are associated with the various services found on modern networks?



How does a web server function?



What are the services utilized by email systems and servers?



How do remote access and monitoring services assist system administrators?

Learning Outcomes

Networked Host Services

File/Print Servers

Database Servers

Web Servers

Hypertext Transfer Protocol Secure

Mail Servers

Mailbox Servers

Directory and Authentication Servers

Remote Terminal Access Servers

Time Servers

Network Monitoring Servers

File/Print Servers

Server Message Block

- Common Internet File System

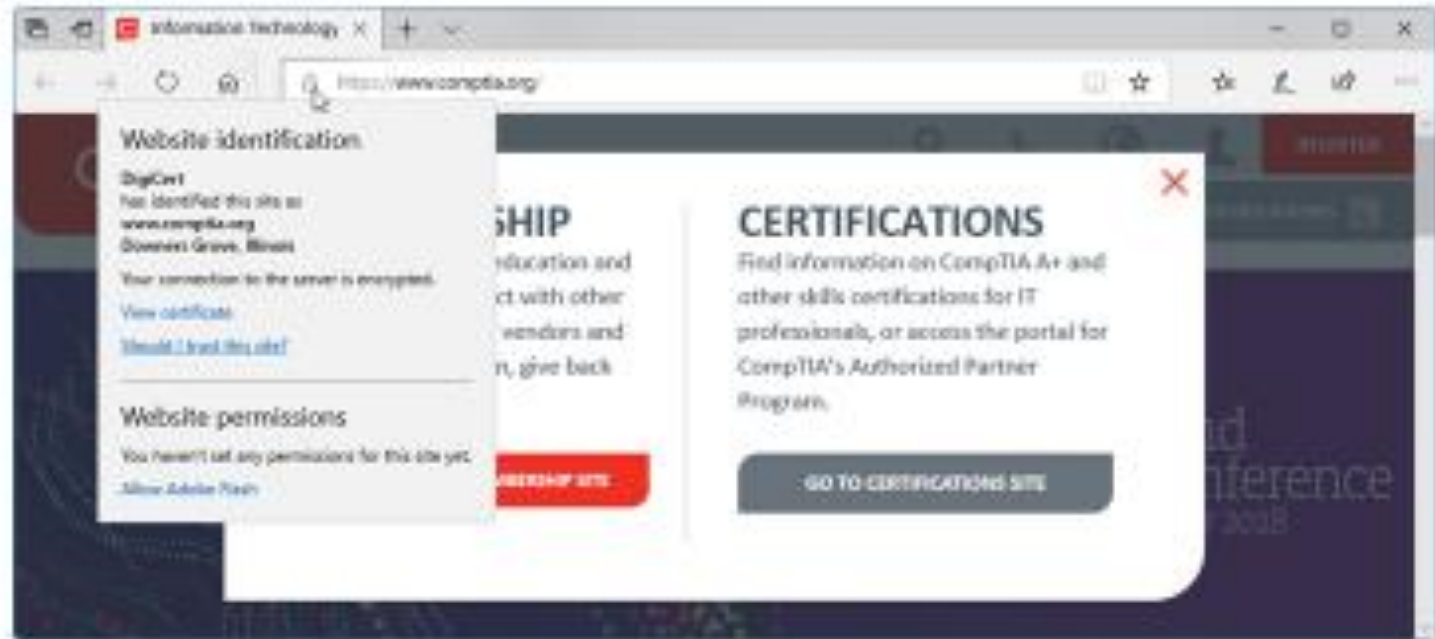
Network Basic Input/Output System

- Earlier Windows used NetBIOS, but now networks use IP, TCP/UDP, and DNS

File Transfer Protocol

- Allows client to upload/download files from network server

HTTPS padlock icon



A server that guarantees subject identities by issuing signed digital certificate wrappers for their public keys.

Database Servers



Store large amounts of structured and unstructured data



Relational databases (SQL)

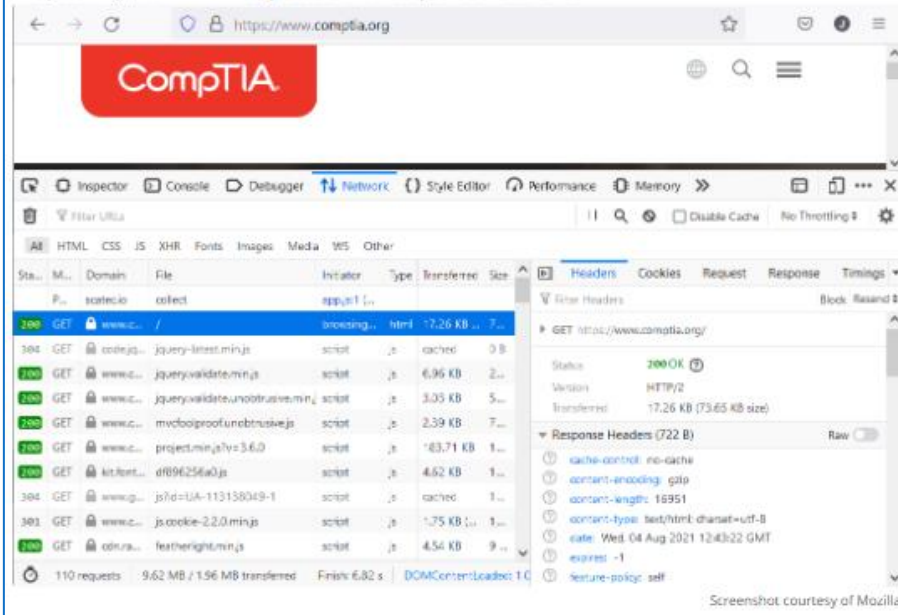
The most used database type
Links data by relationship to one another



Non-relational databases

Store data with flexibility; uses graphs, documents, key-value points

Inspecting the HTTP requests and response headers



The screenshot shows a web browser with the URL <https://www.comptia.org>. The Network tab is open, displaying a list of requests. The first request is a GET request to <https://www.comptia.org/>, which is highlighted. The Headers panel on the right shows the response headers for this request, including Status: 200 OK, Version: HTTP/2, and various cache and content-related headers.

St...	M...	Domain	File	Initiator	Type	Transferre	Size
300	GET	www.c...	/	request...	html	17.26 KB	7...
304	GET	www.c...	jquery-latest.min.js	script	js	cached	0 B
200	GET	www.c...	jquery.validate.min.js	script	js	6.96 KB	2...
200	GET	www.c...	jquery.validate.unobtrusive.min.js	script	js	3.05 KB	5...
200	GET	www.c...	mvdbolproofuncbtrusvejs	script	js	2.39 KB	7...
200	GET	www.c...	project.min.js?v=3.6.0	script	js	163.71 KB	1...
200	GET	kitfont...	d8896256a0.js	script	js	4.62 KB	1...
304	GET	www.c...	js?d=UA-113138049-1	script	js	cached	1...
301	GET	www.c...	js.cookie-2.2.0.min.js	script	js	1.75 KB	1...
200	GET	cdn.ra...	featherlight.min.js	script	js	4.54 KB	9...

110 requests 9.62 MB / 1.96 MB transferred Finish: 6.82 s DOMContentLoaded: 1.0 s

Headers

Request Headers

GET <https://www.comptia.org/>

Status: 200 OK

Version: HTTP/2

Transferred: 17.26 KB (73.65 KB size)

Response Headers (722 B)

cache-control: no-cache

content-encoding: gzip

content-length: 16951

content-type: text/html; charset=utf-8

date: Wed, 04 Aug 2021 12:43:22 GMT

expires: -1

feature-policy: self

Screenshot courtesy of Mozilla.

Web Servers

HyperText Transfer Protocol

Client connects by port TCP/80 to submit resource request

Server returns data or responds with error code

HTML, forms and web applications

Uniform Resource Locators (URLs)

Protocol

FDQN

File path

Hypertext Transfer Protocol Secure

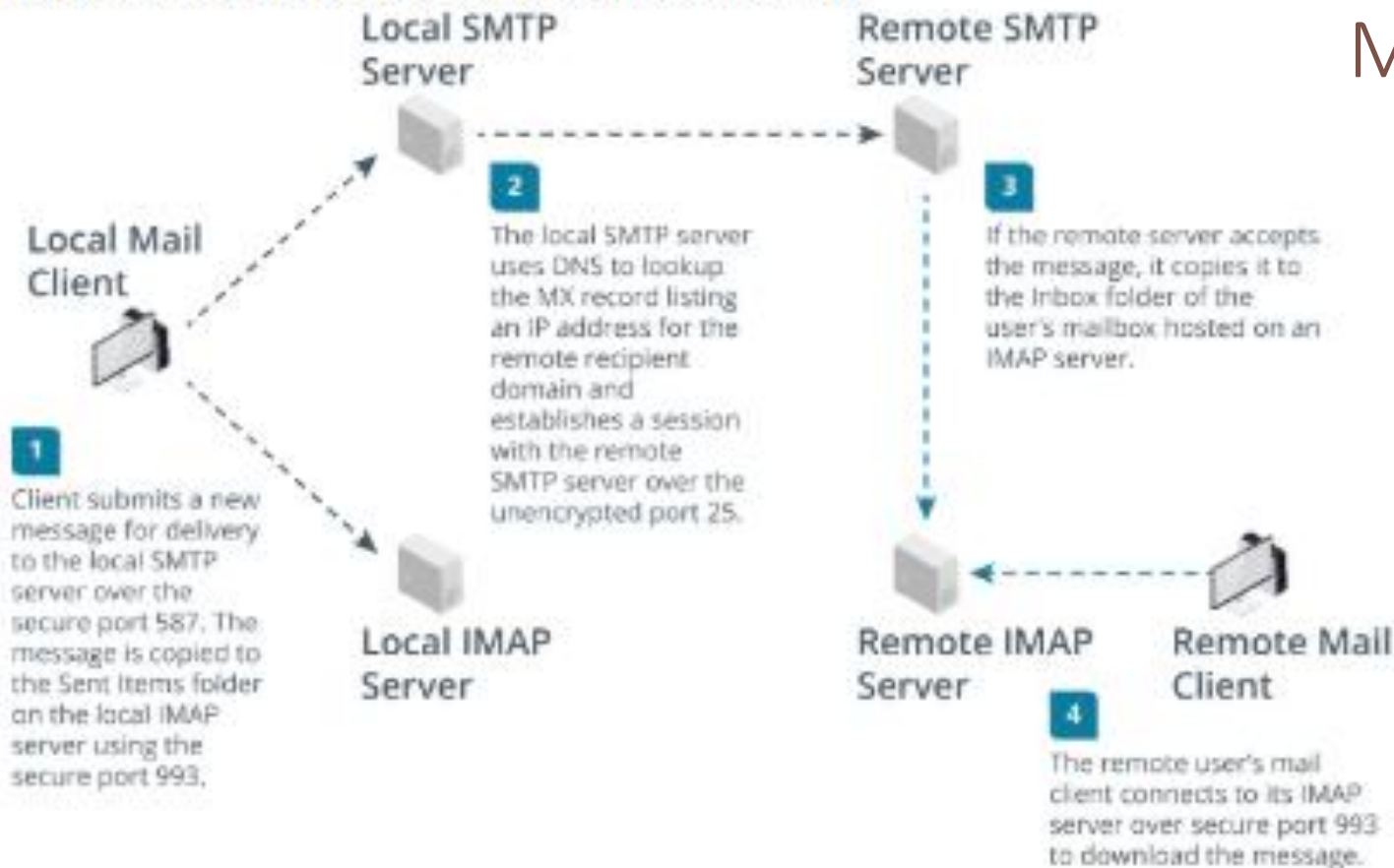
HTTP

- Data is unencrypted, no client/server authentication
- Secure Sockets Layer, and Transport Layer Security were developed to address security

HTTPS

- TLS used with HTTP
- Uses public/private encryption key pair

Operation of delivery and mailbox email protocols



Mail Servers

The two types of servers/protocols for email are mail transfer and mailbox access protocols

Simple Mail Transfer Protocol (SMTP)

Specifies how email is delivered from one domain to another

Uses port TCP/25 and port TCP/587

Mailbox Servers

Post Office Protocol 3 (POP3)

POP client establishes connection on port TCP/110 or secure port TCP/995

Internet Message Access Protocol (IMAP)

Addresses some limitations of POP

Connects multiple clients to mailbox at once

Can organize messages in folders, use multiple mailboxes

The screenshot shows the 'Configuring an email account' window, specifically the 'Add Account' tab. The window is titled 'Configuring an email account' and has a close button in the top right corner. Below the title bar, there is a section for 'POP and IMAP Account Settings' with the instruction 'Enter the mail server settings for your account.' The window is divided into several sections: 'User Information' with fields for 'Your Name' (David Martin) and 'Email Address' (david@davidmartin.me); 'Server Information' with a dropdown for 'Account Type' set to 'POP3', and fields for 'Incoming mail server' (pop3.myisp.net) and 'Outgoing mail server (SMTP)' (smtp.myisp.net); 'Logon Information' with fields for 'User Name' (david) and 'Password' (masked), and checkboxes for 'Remember password' and 'Negotiate logon using Secure Password Authentication (SPA)'; 'Test Account Settings' with a 'Test Account Settings...' button and a checkbox for 'Automatically test account settings when Next is clicked'; and 'Deliver new messages to:' with radio buttons for 'New Outlook Data File' (selected) and 'Existing Outlook Data File', followed by a 'Browse' button. At the bottom right, there is a 'More Settings...' button. At the very bottom of the window, there are navigation buttons: '< Back', 'Next >', and 'Cancel'.

Configuring an email account

Add Account

POP and IMAP Account Settings
Enter the mail server settings for your account.

User Information

Your Name: David Martin

Email Address: david@davidmartin.me

Server Information

Account Type: POP3

Incoming mail server: pop3.myisp.net

Outgoing mail server (SMTP): smtp.myisp.net

Logon Information

User Name: david

Password: [masked]

☒ Remember password

☒ Negotiate logon using Secure Password Authentication (SPA)

Test Account Settings

We recommend that you test your account to ensure that the entries are correct.

Test Account Settings ...

☒ Automatically test account settings when Next is clicked

Deliver new messages to:

☒ New Outlook Data File

☐ Existing Outlook Data File

Browse

More Settings ...

< Back Next > Cancel

Directory and Authentication Servers



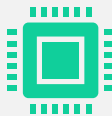
Lightweight
Directory Access
Protocol

TCP/IP protocol
used to query
and update
X.500 directory



Authentication,
Authorization,
Accounting server

Consolidates
authentication
services across
many devices

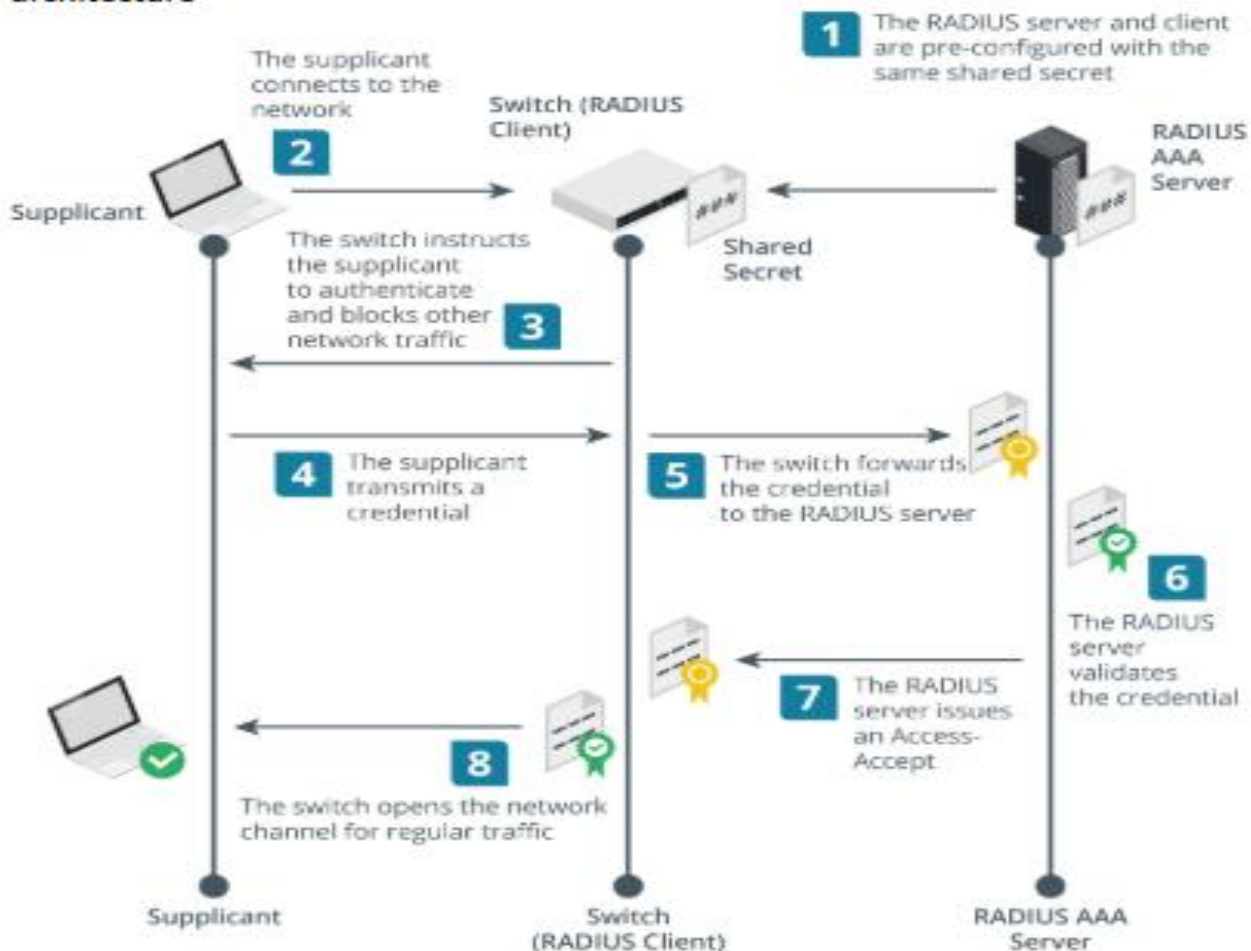


AAA components

Supplicant,
network access
server (NAS), and
AAA server

Communications between RADIUS server, client, and supplicant in AAA architecture

CompTIA



Remote Terminal Access Servers

Teletype device

Secure Shell (SSH)

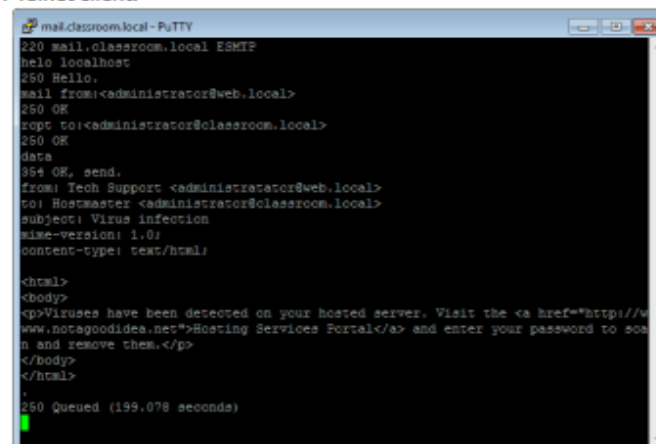
Telnet

Remote Desk
protocol

“Terminal” is from
early days of
configuration by
TTY device

Terminal emulator
is software that
replicates TTY
input and output

PuTTY Telnet client.

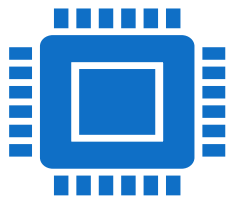


```
mail.classroom.local - PuTTY
220 mail.classroom.local SMTP
helo localhost
250 Hello.
mail from:<administrator@web.local>
250 OK
rcpt to:<administrator@classroom.local>
250 OK
data
354 OK, send.
From: Tech Support <administrator@web.local>
To: Hostmaster <administrator@classroom.local>
Subject: Virus infection
MIME-version: 1.0
content-type: text/html

<html>
<body>
<p>Viruses have been detected on your hosted server. Visit the <a href="http://www.noragoodidea.net">Hosting Services Portal</a> and enter your password to see them and remove them.</p>
</body>
</html>

250 Queued (199.078 seconds)
```


Time Servers



Network Time Protocol (NTP)

Allows systems to sync their clocks to common source

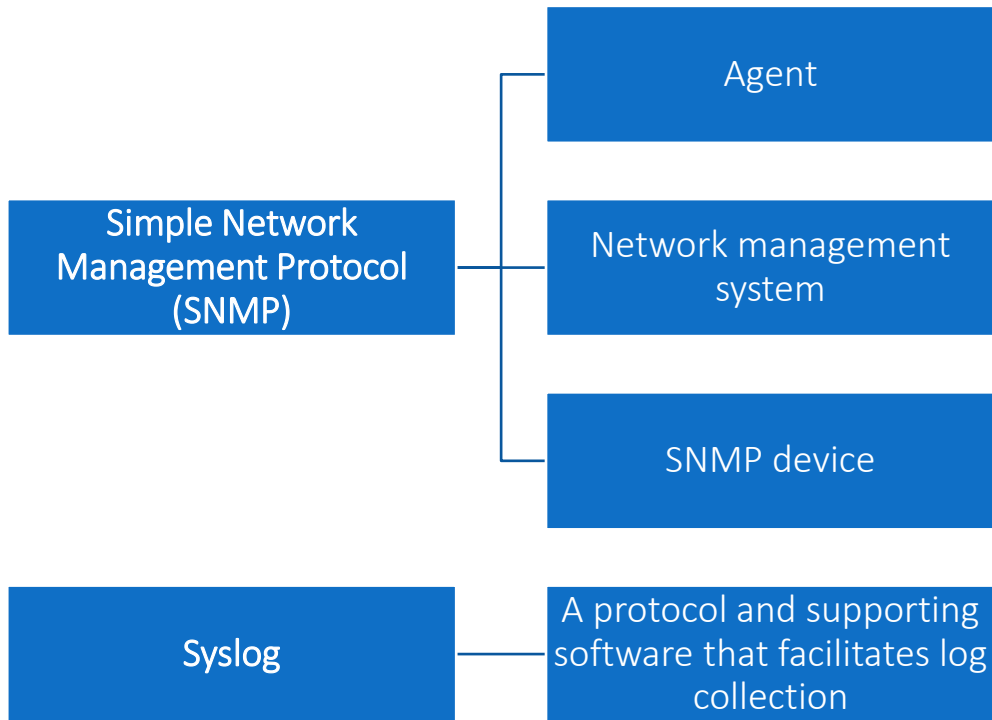


Clock accuracy levels

Described with Stratum levels

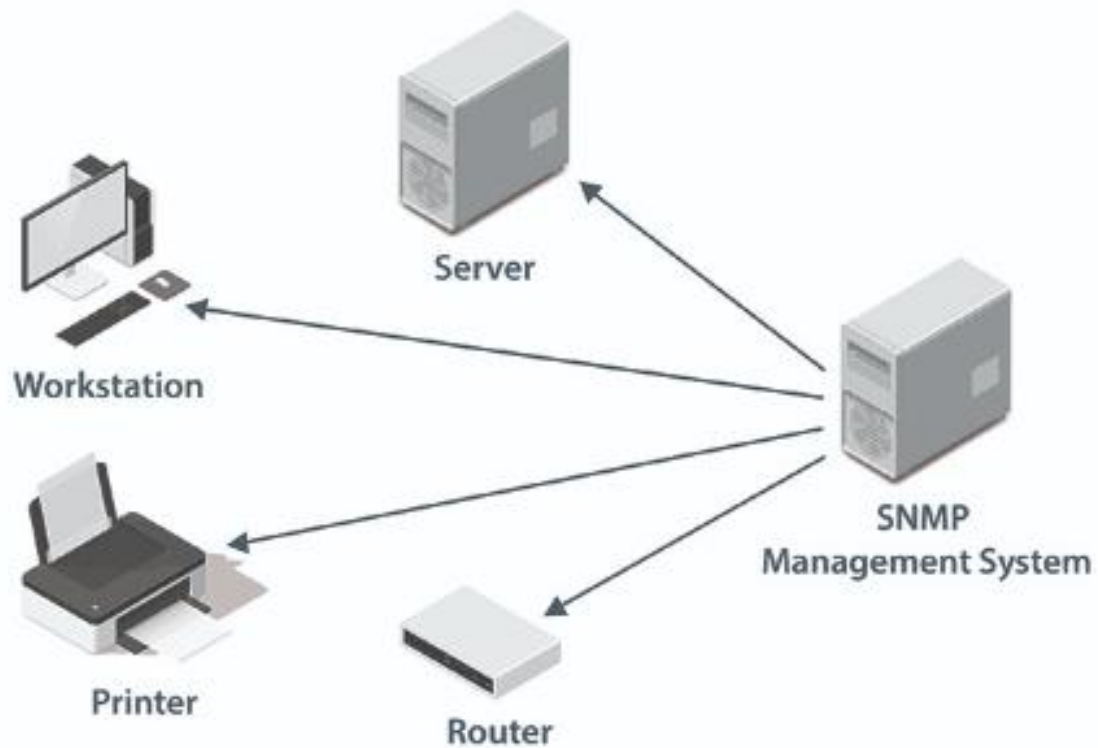
Atomic clock is most accurate (Stratum-0)

NTP servers that sync clock to a Stratum-1 system are a Stratum-2



Network Monitoring Servers

SNMP agents and management system



Configuring an OPNsense security appliance to transmit logs to a remote syslog server

The screenshot shows the OPNsense web interface with the 'Edit destination' dialog box open. The background interface includes a sidebar with navigation options like 'High Availability', 'Routes', 'Settings', 'Administration', 'Cron', 'General', 'Logging', 'Logging / target', 'Miscellaneous', 'Tunables', 'Trust', 'Wizard', 'Log Files', 'Diagnostics', 'Interfaces', 'Firewall', and 'VPN'. The top bar shows the user 'root@OPNsense.localdomain' and a search bar.

Edit destination

Full help ⓘ

Enabled ☒

Transport UDP(4) [Clear All](#)

Applications Nothing selected [Clear All](#)

Levels info, notice, warn, error, critical, alert, emergency [Clear All](#)

Facilities Nothing selected [Clear All](#)

Hostname

Port 514

Lesson Review: Networked Host Services

- File/Print Servers
- Database Servers
- Web Servers
- Hypertext Transfer Protocol Secure
- Mail Servers
- Mailbox Servers
- Directory and Authentication Servers
- Remote Terminal Access Servers
- Time Servers
- Network Monitoring Servers

Lesson 7.2

Internet and Embedded Appliances

Learning Outcomes



How does a proxy server work?



How does a spam gateway function?



How does a load balancer increase the availability of resources?



What is the purpose of a unified threat management system?



What are several examples of IoT devices?

Internet and Embedded Appliances

Proxy Servers

Spam Gateways
and Unified Threat
Management

Load Balancers

Legacy Systems

Embedded
Systems and
SCADA

Internet of Things
Devices

Proxy Servers

Functions

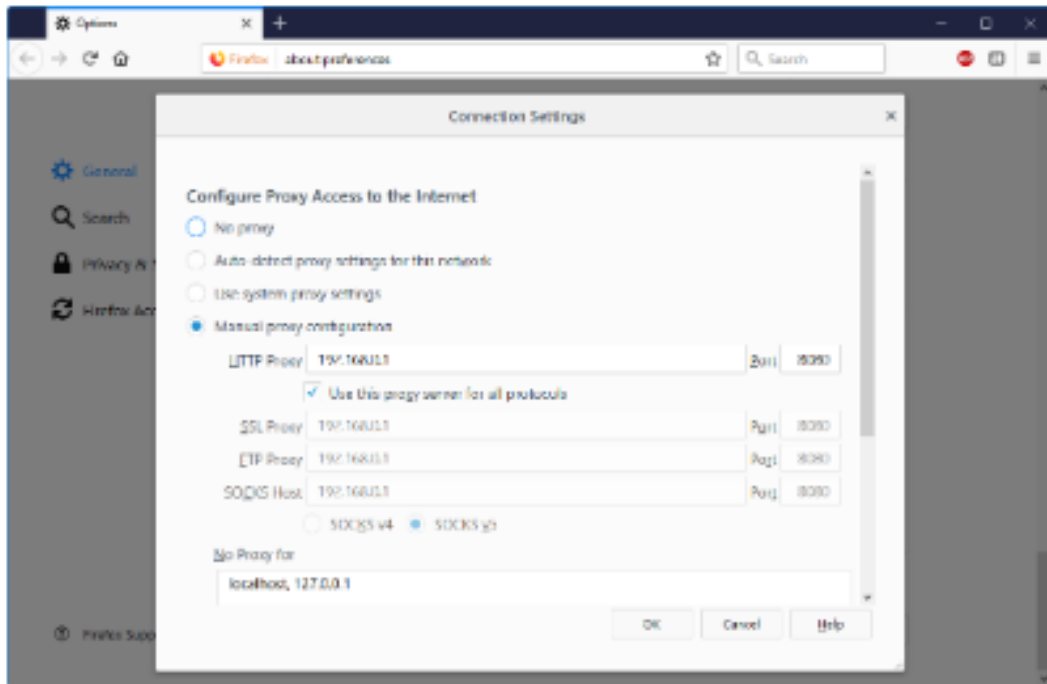
Takes a whole HTTP request from client, checks it, then forwards to server

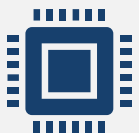
Checks reply then shuttles it back to LAN computer

Operates as transparent service or nontransparent proxy

Provides security by blocking access to sites deemed inappropriate

Configuring the Firefox web browser to use a proxy server at 192.168.0.1 to connect to the Internet





Range of functions

Firewalls

Intrusion detection system (IDS)

Antivirus/antimalware solutions

Spam gateways

Content filters

Data leak/loss prevention (DLP)



Unified threat management (UTM)

Spam Gateways and Unified Threat Management

Load Balancers



USED WITH MULTIPLE SERVERS
(WEB, EMAIL, STREAMING)



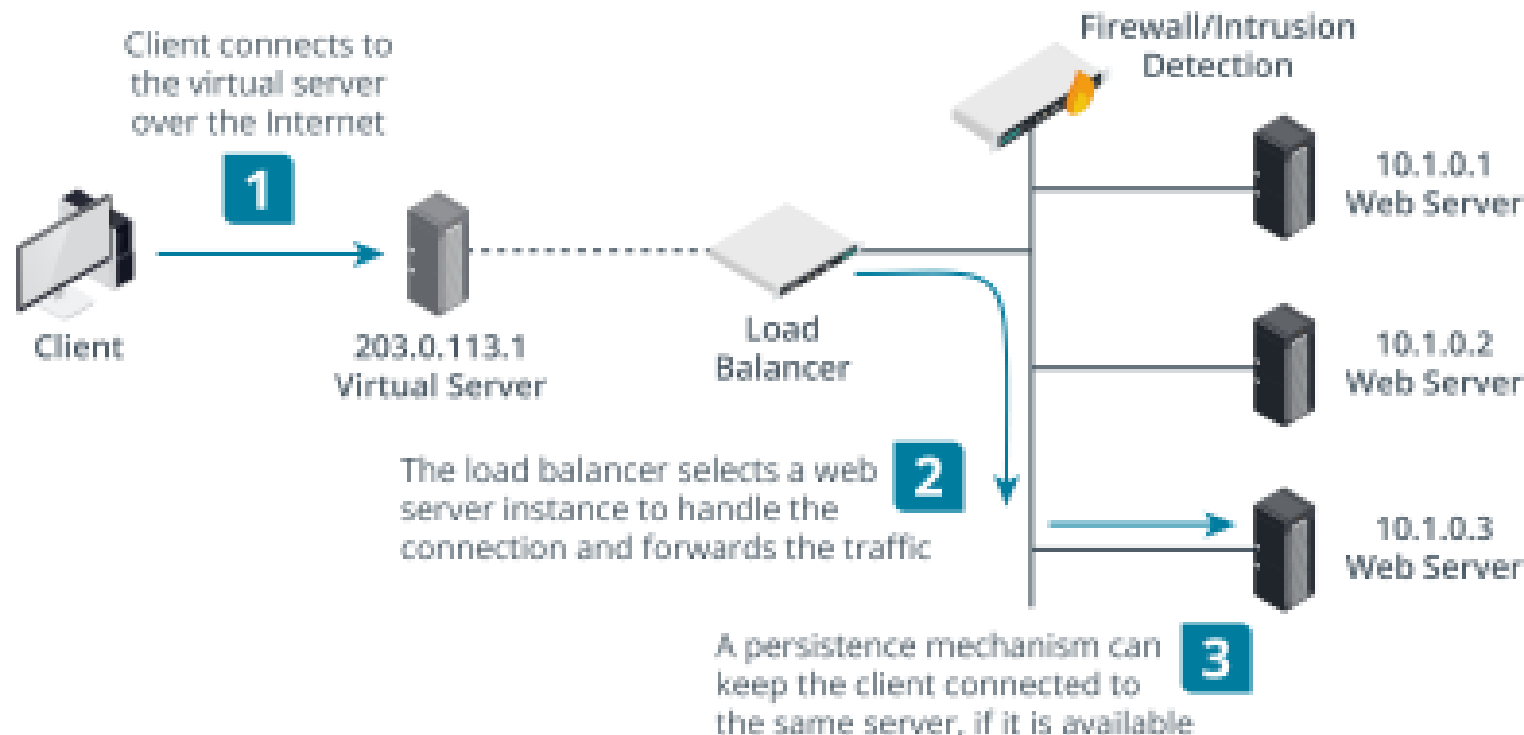
PLACED IN FRONT OF SERVER
NETWORK



DISTRIBUTES REQUESTS FROM
CLIENT NETWORK OR INTERNET
TO APPLICATION SERVERS

Load Balancers

Topology of basic load balancing architecture.



Legacy Systems

End-of-life (EOL) systems no longer supported by vendor

Networks retain them to run services that are complex or costly to migrate

Security risks since vendor no longer develops software patch to block attacks

Embedded Systems and SCADA

– Supervisory control and data acquisition

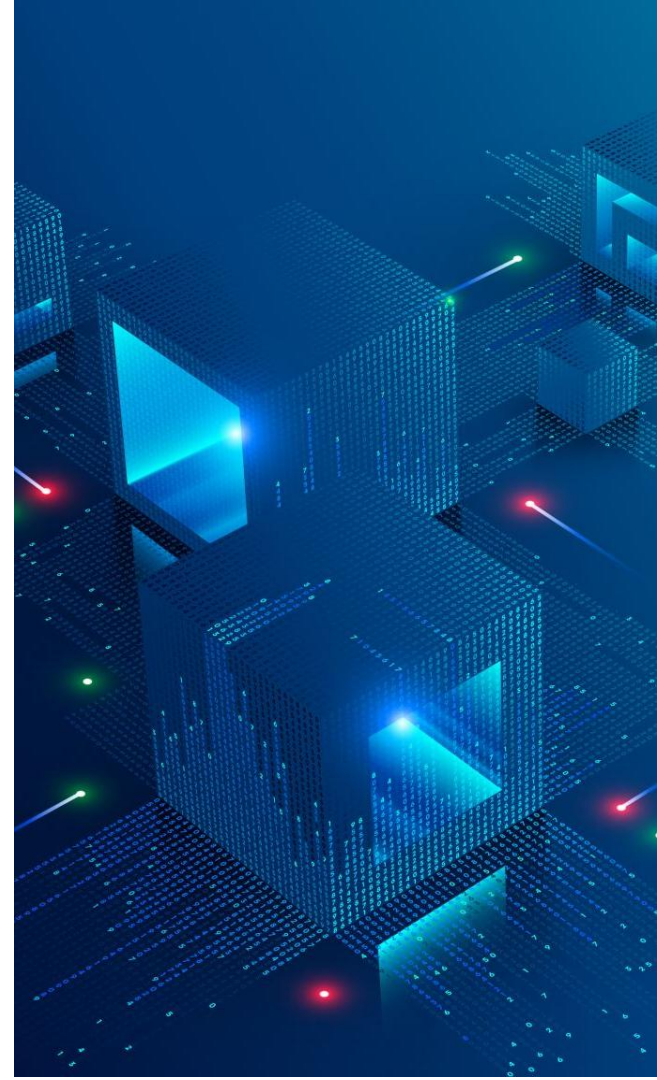
Embedded system

Device designed to perform dedicated function

Referred to as operational technology network

Workflow and process automation systems

Industrial control system (ICS)



Internet of Things Devices

Global network of devices that share data and services over Internet

IoT smart device networks

Hub/control system (i.e., smart speaker operated by voice)

Smart device (i.e., lightbulb, refrigerator, thermostat)



Lab Activity

- Lab: Use a Proxy Server
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Internet and Embedded Appliances

- Proxy Servers
- Spam Gateways and Unified Threat Management
- Load Balancers
- Legacy Systems
- Embedded Systems and SCADA
- Internet of Things Devices

Lesson 7.3

Troubleshoot Networks

A+ 220-1201 Certification Exam Objective:

- 5.5 Given a scenario, troubleshoot network issues.

Learning Outcomes



What are common issues found on wired networks and how are they resolved?



What are the common issues found in wireless networks and how are they resolved?



What unique problems stem from the use of VoIP systems on a network?

Troubleshoot Networks

01

Troubleshoot
Wired
Connectivity

02

Troubleshoot
Network
Speed Issues

03

Troubleshoot
Wireless
Issues

04

Troubleshoot
VoIP Issues

05

Troubleshoot
Limited
Connectivity

Troubleshoot Wired Connectivity

Components are NIC port, RJ45 patch cords, structured data cable, network transceiver

Troubleshooting steps

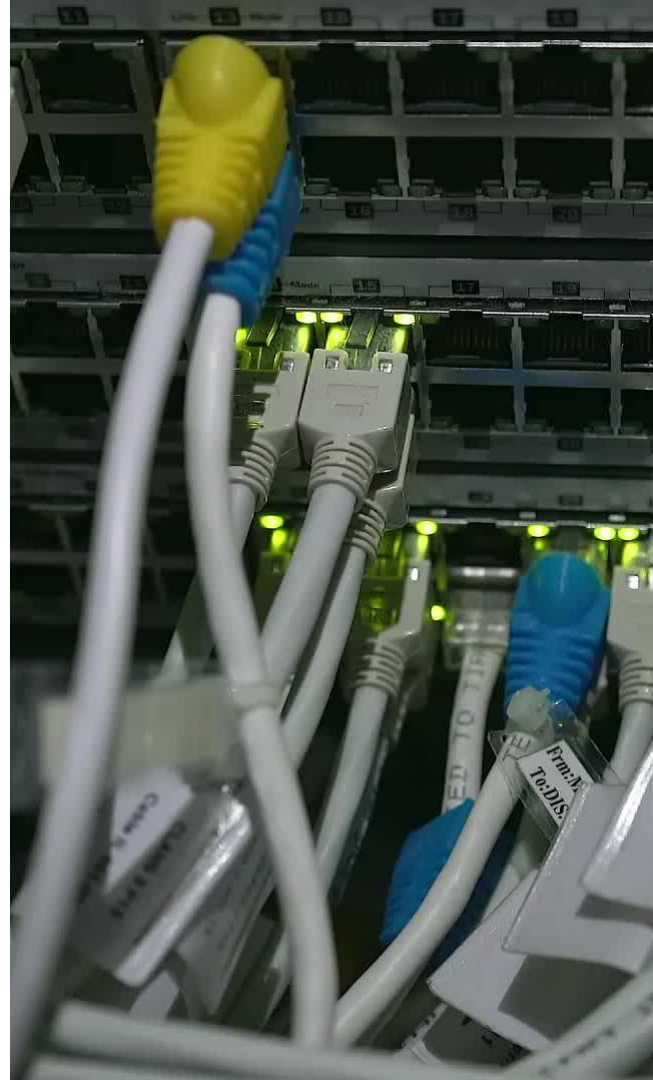
- Check patch cords are terminated/connected

- Test transceivers using loopback tool

- Use cable tester to verify structured cabling

- Verify Ethernet speed/duplex configuration on switch/NIC

Troubleshoot port flapping issues (up & down)



Troubleshoot Network Speed Issues

Troubleshooting steps

Identify network **activity** **user** is performing

Isolate the speed issue to a **single cable segment**

Install **update** for network **adapter driver**

Consider that computer could be infected with **malware**

Establish problem scope; is it a **single user** or **all users**



Troubleshoot Wireless Issues

Possible causes

Received signal
strength indicator

Troubleshooting
wireless signal issues

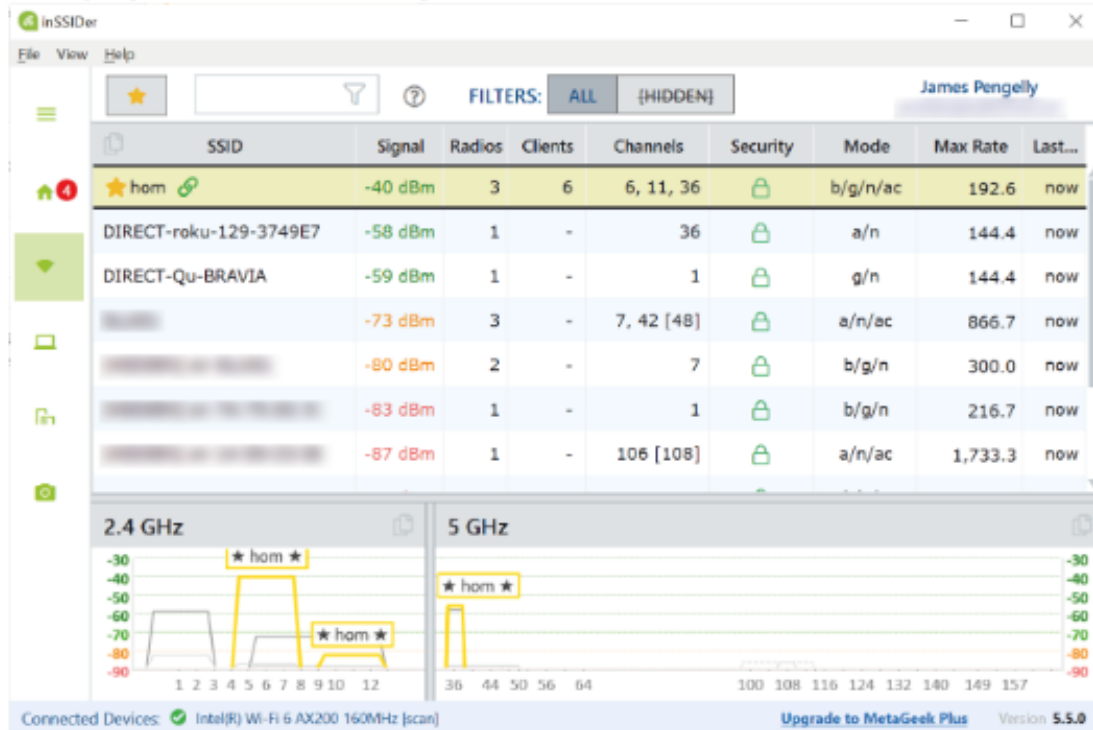
Move devices closer
together

User may be out of
range or SSID broadcast
suppressed

Mismatch of standards,
or older device slows
down whole network

Some clients
supporting 802.11n
may not have dual-
band radios

Surveying Wi-Fi networks using inSSIDer



Troubleshoot
Wireless Issues

Troubleshoot VoIP Issues

Uses data networks for voice calling

Problems with timing/sequence of packet delivery

- **Latency:** time for signal to reach recipient (in ms)
- **Jitter:** amount of variation in delay over time

VoIP quality is only established by using quality of service (QoS)

Troubleshoot Limited Connectivity

Limited connectivity

- Host has physical connection, but no lease for an IP configuration from DHCP server

Troubleshooting steps

- Establish scope of the issue
- Check configuration of patch cords
- Check the VLAN configuration

Lab Activity

- Lab: Fix a Network Connection
(available in CertMaster Learn and CertMaster Perform)
- Lab: Troubleshoot a Network Issue
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Troubleshoot Networks

- Troubleshoot Wired Connectivity
- Troubleshoot Network Speed Issues
- Troubleshoot Wireless Issues
- Troubleshoot VoIP Issues
- Troubleshoot Limited Connectivity